EE1 - SISTEMAS DIGITAIS (2026.1)

Questão 1.

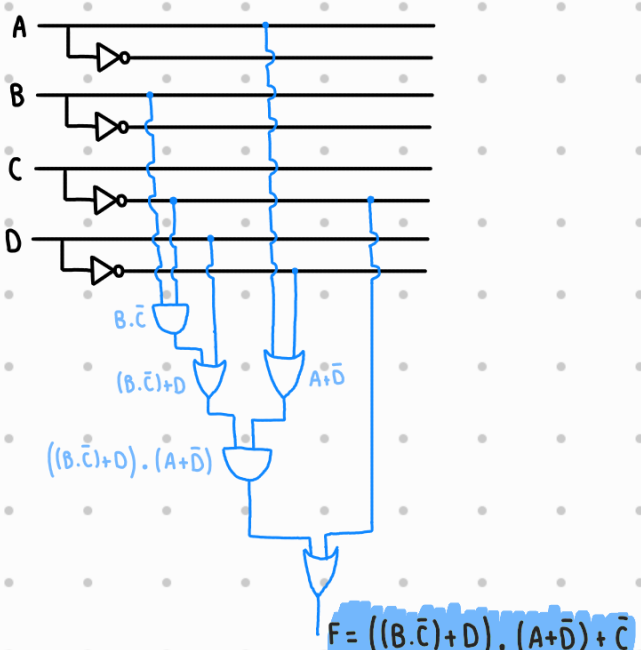
$$F = \underbrace{((B.\bar{C}) + D).(A + \bar{D})}_{Z} + \bar{C}$$

a) Tabela-verdade: $F = Z + \bar{C}$

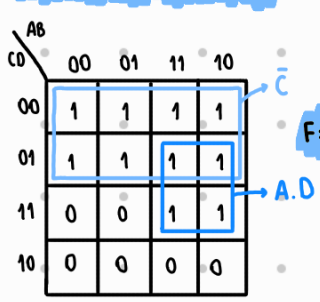
OBS: Sempre que $C=0$ ($\bar{C}=1$), sempre $F=1$; pois $x+1=1$

A	B	C	D	Z	$\bar{C}$	F
0	0	0	0	0	1	1
0	0	0	1	0	1	1
0	0	1	0	0	0	0
0	0	1	1	0	0	0
0	1	0	0	1	1	1
0	1	0	1	0	1	1
0	1	1	0	0	0	0
0	1	1	1	0	0	0
1	0	0	0	0	1	1
1	0	0	1	1	1	1
1	0	1	0	0	0	0
1	0	1	1	1	0	1
1	1	0	0	1	1	1
1	1	0	1	1	1	1
1	1	1	0	0	0	0
1	1	1	1	1	0	1

Circuito lógico: (sem otimizações e com barramento)

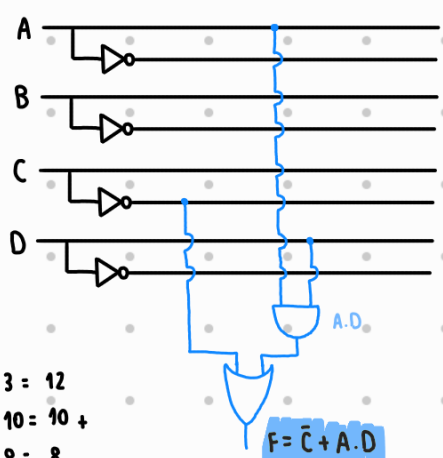


b) Mapa de Karnaugh:



$F = \bar{C} + A.D$

Circuito digital reduzido:



Área total dos circuitos:

a) 4 portas NOT: $4 \cdot 3 = 12$
2 portas AND: $2 \cdot 10 = 20$ +
3 portas OR: $3 \cdot 8 = 24$
56 mm²

b) 4 portas NOT: $4 \cdot 3 = 12$
1 porta AND: $1 \cdot 10 = 10$ +
1 porta OR: $1 \cdot 8 = 8$
30 mm²

Questão 2.

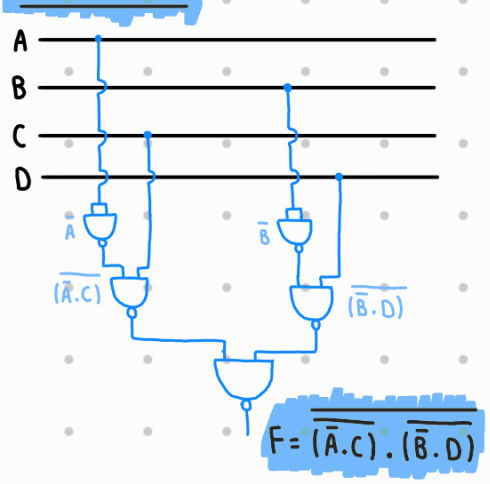
O objetivo é transformar todas as operações OR em AND

$F = (\bar{A}.C) + \overline{(B + \bar{D})}$ — aplicando De Morgan: $\overline{(B+\bar{D})} = \bar{B}.D$

$F = (\bar{A}.C) + (\bar{B}.D)$ — aplicando De Morgan: $(\bar{A}.C) + (\bar{B}.D) = \overline{(\bar{A}.C)}.\overline{(\bar{B}.D)}$

$$F = \overline{(\bar{A}.C)}.\overline{(\bar{B}.D)}$$

Circuito digital: (utilizando apenas portas NAND)



Questão 3. $R = 3X + 2Y$

a) Tabela-verdade:

X_1	X_0	Y_1	Y_0	$R = 3X + 2Y$	R_4	R_3	R_2	R_1	R_0
0	0	0	0	3.0 + 2.0 = 0	0	0	0	0	0
0	0	0	1	3.0 + 2.1 = 2	0	0	0	1	0
0	0	1	0	3.0 + 2.2 = 4	0	0	1	0	0
0	0	1	1	3.0 + 2.3 = 6	0	0	1	1	0
0	1	0	0	3.1 + 2.0 = 3	0	0	0	1	1
0	1	0	1	3.1 + 2.1 = 5	0	0	1	0	1
0	1	1	0	3.1 + 2.2 = 7	0	0	1	1	1
0	1	1	1	3.1 + 2.3 = 9	0	1	0	0	1
1	0	0	0	3.2 + 2.0 = 6	0	0	1	1	0
1	0	0	1	3.2 + 2.1 = 8	0	1	0	0	0
1	0	1	0	3.2 + 2.2 = 10	0	1	0	1	0
1	0	1	1	3.2 + 2.3 = 12	0	1	1	0	0
1	1	0	0	3.3 + 2.0 = 9	0	1	0	0	1
1	1	0	1	3.3 + 2.1 = 11	0	1	0	1	1
1	1	1	0	3.3 + 2.2 = 13	0	1	1	0	1
1	1	1	1	3.3 + 2.3 = 15	0	1	1	1	1

b) mapa-K de R_4:

$R_4 = 0$

Y_1Y_0 \ X_1X_0	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	0	0
10	0	0	0	0

mapa-K de R_3:

$R_3 = X_1 \cdot Y_0 + X_1 \cdot Y_1 + X_1 \cdot X_0 + X_0 \cdot Y_1 \cdot Y_0$

Y_1Y_0 \ X_1X_0	00	01	11	10
00	0	0	1	0
01	0	0	1	1
11	0	1	1	1
10	0	0	1	1

$X_1 \cdot Y_0$; $X_1 \cdot Y_1$; $X_1 \cdot X_0$; $X_0 \cdot Y_1 \cdot Y_0$

mapa-K de R_1:

$R_1 = X_1 \cdot X_0 \cdot Y_0 + X_1 \cdot \overline{X_0} \cdot \overline{Y_0} + \overline{X_1} \cdot \overline{X_0} \cdot Y_0 + \overline{X_1} \cdot X_0 \cdot \overline{Y_0}$

Y_1Y_0 \ X_1X_0	00	01	11	10
00	0	1	0	1
01	1	0	1	0
11	1	0	1	0
10	0	1	0	1

$X_1 \cdot X_0 \cdot Y_0$; $X_1 \cdot \overline{X_0} \cdot \overline{Y_0}$; $\overline{X_0} \cdot \overline{X_1} \cdot Y_0$; $\overline{X_1} \cdot X_0 \cdot \overline{Y_0}$

mapa-K de R_2:

$R_2 = X_0 \cdot Y_1 \cdot \overline{Y_0} + X_1 \cdot Y_1 \cdot Y_0 + \overline{X_1} \cdot \overline{X_0} \cdot Y_1 + X_1 \cdot \overline{X_0} \cdot \overline{Y_1} \cdot \overline{Y_0} + \overline{X_1} \cdot X_0 \cdot \overline{Y_1} \cdot Y_0$

Y_1Y_0 \ X_1X_0	00	01	11	10
00	0	0	0	1
01	0	1	0	0
11	1	0	1	1
10	1	1	1	0

$\overline{X_1} \cdot X_0 \cdot \overline{Y_1} \cdot Y_0$; $X_1 \cdot \overline{X_0} \cdot \overline{Y_1} \cdot \overline{Y_0}$; $X_1 \cdot Y_1 \cdot Y_0$; $X_0 \cdot Y_1 \cdot \overline{Y_0}$; $\overline{X_0} \cdot \overline{X_1} \cdot Y_1$

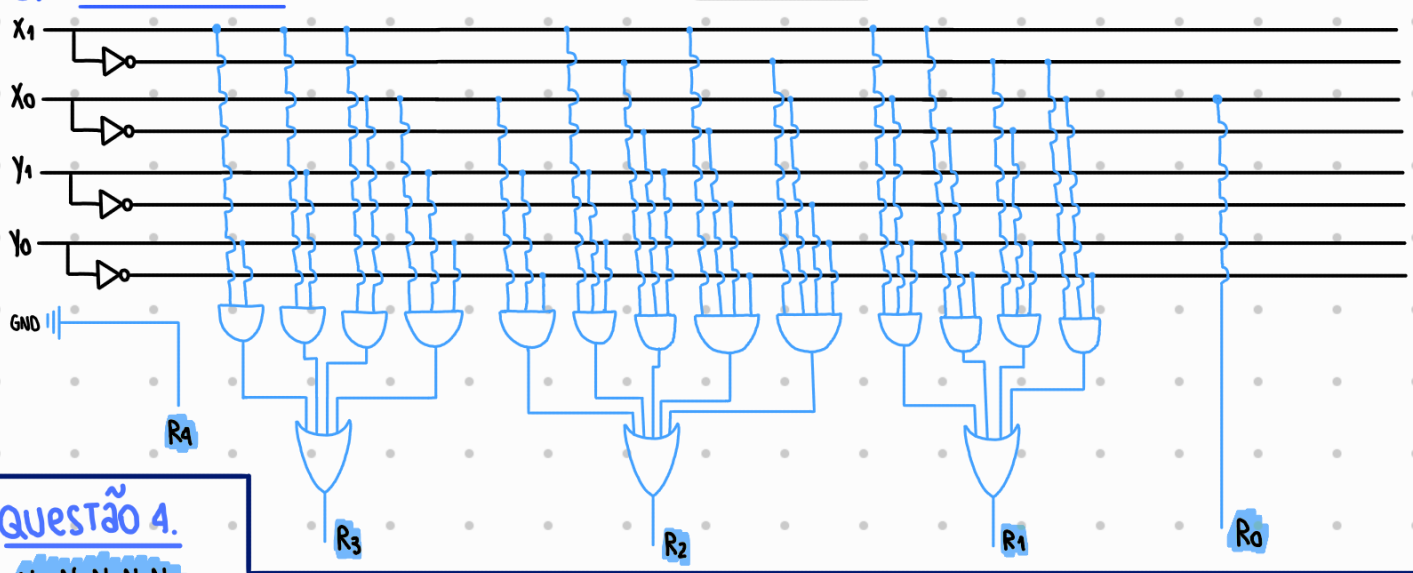
mapa-K de R_0:

$R_0 = X_0$

Y_1Y_0 \ X_1X_0	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	0	1	1	0
10	0	1	1	0

X_0

c) Circuito digital:



Questão 4.

$N = N_3N_2N_1N_0$

Tabela-verdade:

N_3	N_2	N_1	N_0	F	G
0	0	0	0	1	0
0	0	0	1	0	0
0	0	1	0	0	1
0	0	1	1	0	1
0	1	0	0	0	0
0	1	0	1	1	1
0	1	1	0	0	0
0	1	1	1	0	1
1	0	0	0	0	0
1	0	0	1	0	0
1	0	1	0	1	0
1	0	1	1	0	1
1	1	0	0	0	0
1	1	0	1	0	1
1	1	1	0	0	0
1	1	1	1	1	0

mapa-K de F:

$F = \overline{N_3}\overline{N_2}\overline{N_1}\overline{N_0} + \overline{N_3}N_2\overline{N_1}N_0 + N_3N_2N_1N_0 + N_3\overline{N_2}N_1\overline{N_0}$

N_1N_0 \ N_3N_2	00	01	11	10
00	1	0	0	0
01	0	1	0	0
11	0	0	1	0
10	0	0	0	1

mapa-K de G:

$G = N_2\overline{N_1}N_0 + \overline{N_3}N_1N_0 + \overline{N_2}N_1N_0 + N_3\overline{N_2}N_1$

N_1N_0 \ N_3N_2	00	01	11	10
00	0	0	0	0
01	0	1	1	0
11	1	1	0	1
10	1	0	0	0

Circuito digital:

